Learn

Completed

Understanding LMS

LMS stands for **Learning Management System**.

In simple words, an LMS is a platform where:

- Instructors organize learning content
- Students enroll in a course
- Lessons are arranged in order
- Videos are delivered in sequence
- Student progress is tracked

Where Is LMS Used?

LMS platforms are commonly used in:

- Colleges and universities
- Online coaching platforms
- Corporate training programs
- Skill development platforms
- Internal company learning portals

Anywhere structured learning is needed, an LMS is used.

From Student Perspective (User View)

Let's think like a student 👩‍🎓

When you open a LMS that uses YouTube videos, what do you expect?

Course Listing Page

- List of courses (Java, Python, ML, etc.)
- Course thumbnail
- Instructor name
- Short description
- Enroll button

Course Details Page

- Course description
- What you will learn
- Total number of lessons
- Total duration
- Enroll button

Learning Page (Main Area)

This is the most important part.

Usually it contains:

- 🎥 Video player on top (YouTube embedded player)
- 📚 Lesson list on the side
- ✅ Progress indicator
- ⏭️ Next / Previous buttons

Student flow:

- Click lesson → YouTube video loads
- Finish video → System marks lesson as completed
- Move to next lesson

Even though video comes from YouTube, the learning structure is controlled by your LMS.

Progress Tracking

- Percentage completed
- Completed lessons marked
- Resume from last watched lesson

How LMS Works

You already know:

- Videos are hosted on YouTube(or S3 or any other storage)
- Only YouTube video URLs (or video IDs) are stored in database

Now let's understand full flow in simple terms.

Step 1: Student Opens Course

Frontend → sends request to backend:

"Give me all lessons for Java course"

Backend:

- Fetches lessons from database
- Each lesson contains:
  - Title
  - Order number
  - YouTube URL (or video ID)

Frontend:

- Displays lesson list
- Loads first YouTube video inside iframe

Student Clicks a Lesson

Frontend:

- Sends lesson ID to backend

Backend:

- Returns corresponding YouTube URL

Frontend:

- Updates iframe source
- YouTube player loads that video

> **Important: Video is not stored in your system. You are only embedding YouTube video using its URL.**

Progress Update

When student completes a lesson:

Frontend:

- Sends "lesson completed" to backend

Backend:

- Stores:
  - user_id
  - course_id
  - lesson_id
  - status = completed

> **Next time: Backend calculates progress like:**
> **completed lessons ÷ total lessons × 100**

Main Modules of LMS

Authentication Module

- Signup
- Login
- Role (Student / Instructor / Admin)

Course Module

Stores:

- Course title
- Description
- Thumbnail
- Category
- Instructor ID

Section & Lesson Module

Structure:

Course
→ Sections
→ Lessons
→ YouTube URL

Example:

Java Course
→ Basics
→ Variables
→ Data Types
→ OOP

Each lesson stores:

- Title
- Order
- YouTube URL
- Duration

Enrollment Module

Tracks:

- Which student enrolled in which course
- Enrollment date

Progress Tracking Module

Tracks:

- Completed lessons
- Percentage
- Last watched lesson

From Developer Perspective

Now let's think like a developer building this LMS using YouTube videos.

Frontend

Built using:

- React / Angular / Vue

Responsibilities:

- Display courses
- Display lesson list
- Embed YouTube video using iframe
- Track user progress
- Send API requests

Frontend loads video like:

```
https://www.youtube.com/embed/{video_id}
```

Backend

Built using:

- Node / Java / Python

Responsibilities:

- Authentication
- Authorization
- Course management
- Enrollment handling
- Progress tracking
- Returning lesson data

Backend stores:

- Metadata only
- YouTube video URL
- No video files

Database

Typical tables:

- Users
- Courses
- Sections
- Lessons
- Enrollments
- Progress

Lesson table stores:

- lesson_id
- title
- order_number
- youtube_url

Architecture Flow (Simple Version)

```
Student (Browser)
      ↓
Frontend (React)
      ↓ API Call
Backend (Server)
      ↓
Database (Fetch lessons + YouTube URLs)
      ↓
Return data
      ↓
Frontend embeds YouTube video
      ↓
Student watches video from YouTube
```

Important:

- Backend controls structure
- Database controls order
- YouTube only hosts the video
- Your LMS controls learning flow

Final Understanding in One Line

An LMS is:

> A structured system that organizes video-based learning into courses, manages users and enrollments, tracks progress, and controls learning flow.